

Basic Guide to Radio-Controlled Boat Modelling



T.K. Hareendran is an electronics hobbyist, freelance technical writer and circuit designer

Whether you crave action or relaxation, you will relish radio-controlled (RC) boating because there is nothing quite as comforting as gliding your own RC boat across a pond. If you enjoy building and racing RC boats but do not know where to start, read on for an avalanche of helpful hints.

The basics

Easy-to-use boat kits are widely available for the craftsman who wants building satisfaction. Most of the kits may be 90 per cent pre-built and may include an electric motor/engine, radio, radio gear, batteries and other required accessories. Boat kits with a single motor and battery (brushless electric motor and a rechargeable Ni-MH/lithium-polymer battery) are the norm. However, twinmotored boat kits are also available. Take note, the standard running time of a single-motor boat is around three to six minutes only.

Fig. 1: A mini RC boat



Fig. 2: Layout of the basic components of the RC boat



The drive system of a typical RC boat is very simple. An electric motor near the front of the boat is connected to a shaft that runs towards the back of the boat and out through the bottom of the hull. This shaft is contained within a tube, called stuffing tube, which is filled with grease.

The grease in the stuffing tube provides lubrication and also prevents water from entering the hull. At the end of the shaft is the propeller. Steering is accomplished via a completely submerged rudder placed just behind the propeller. A shaft on the rudder protrudes into the hull, where a small servo actuates it in either direction. The radio receiver and electronic speed control (ESC) for the motor are usually enclosed in a single waterproof casing. Emerging from this casing are the radio antenna, rudder servo cable, battery wires and power on/off switch. The built-in battery pack (Ni-MH or lithium-polymer) with at least 1100mAh capacity provides a nominal DC supply voltage of 7.2V (1.2Vx6 Ni-MH or 2C lithium-polymer).

The electronics

RC boats provide a totally different experience than flying a plane or driving a car on many different levels. Do-it-yourself (DIY) kits of ready-to-run/ready-to-float (RTR/RTF) boats have made it easier for just about anyone to get into RC boat modelling easily and quickly. RC boat kits are more popular than ever and for many great reasons. From the fit and finish to the outstanding performance, the quality of today's DIY kits are better than ever. However, besides the aesthetics of the RC boat, there are other considerations to think about such as the power system, drive system, radio system, battery, charger and a lot more.

Power system. A major part in the power system of an RC boat is the brushed or brushless electric motor. While the brushed